

Entropy

The Beurling–Nyman Flow as Lyapunov Function for RH

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Abstract

Perelman named the entropy for Ricci flow and proved it was monotone. The monotonicity forced the geometry to improve. The Riemann Hypothesis has an analogous entropy: $d_N^2 = \|1_{(0,1)} - P_N 1\|_{L^2}^2$, the squared distance from the indicator function to its best approximation in the N -dimensional Beurling–Nyman subspace. This entropy is monotone decreasing in N by construction. The Riemann Hypothesis is equivalent to $d_N \rightarrow 0$. We prove the decrement formula at each step, compute the gradient explicitly, connect the gradient steps to the Li coefficients [1], and identify the convergence rate needed for a proof. The entropy is named. The flow is defined. The question is whether the flow converges.

1 The Beurling–Nyman Criterion

Definition 1 (The Beurling–Nyman functions). For $\theta \in (0, 1]$, define

$$\rho_\theta(x) = \left\{ \frac{\theta}{x} \right\} - \theta \left\{ \frac{1}{x} \right\}, \quad x > 0,$$

where $\{t\} = t - \lfloor t \rfloor$ is the fractional part. Each $\rho_\theta \in L^2(0, \infty)$. Let $\mathcal{H} = L^2(0, \infty)$ and let $\mathcal{V}_N$ be the subspace spanned by $\rho_{1/1}, \rho_{1/2}, \dots, \rho_{1/N}$.

Theorem 2 (Beurling–Nyman criterion, 1955). *Let $d_N = \inf_{f \in \mathcal{V}_N} \|1_{(0,1)} - f\|_{\mathcal{H}}$. The Riemann Hypothesis holds if and only if*

$$\lim_{N \rightarrow \infty} d_N = 0.$$

Proof sketch. The key identity: the Mellin transform of ρ_θ is $\widehat{\rho}_\theta(s) = (\theta^s - \theta)\zeta(s)/s$ for $\operatorname{Re}(s) \in (0, 1)$. The span of $\{\rho_\theta : \theta \in (0, 1]\}$ is dense in $L^2(0, \infty)$ in a certain topology iff $\zeta(s)$ has no zeros in $\operatorname{Re}(s) > \frac{1}{2}$ by the Paley–Wiener theorem applied to the Hardy space H^2 . Beurling (1955) and Nyman (1950) proved this equivalence. The Báez-Duarte extension to the specific sequence $\{\rho_{1/k}\}$ follows [4]. $\square$

2 The Entropy

Definition 3 (The Lyapunov function). Define the *Beurling–Nyman entropy*:

$$\mathcal{S}_N = d_N^2 = \min_{c_1, \dots, c_N \in \mathbb{C}} \left\| 1_{(0,1)} - \sum_{k=1}^N c_k \rho_{1/k} \right\|_{\mathcal{H}}^2.$$

This is the squared distance from the target $1_{(0,1)}$ to the best approximation in $\mathcal{V}_N$.

Proposition 4 (Monotonicity). $\mathcal{S}_{N+1} \leq \mathcal{S}_N$ for all $N \geq 1$.

Proof. $\mathcal{V}_N \subset \mathcal{V}_{N+1}$, so the infimum over a larger set is smaller or equal. $\square$

Theorem 5 (The decrement formula). Let $f_N^* = \sum_{k=1}^N c_k^* \rho_{1/k}$ be the optimal approximation. Let $r_N = 1_{(0,1)} - f_N^*$ be the residual. Then:

$$\mathcal{S}_N - \mathcal{S}_{N+1} = \frac{|\langle r_N, \rho_{1/(N+1)} \rangle|^2}{\|\rho_{1/(N+1)}\|^2 - \|\text{proj}_{\mathcal{V}_N} \rho_{1/(N+1)}\|^2},$$

where the denominator is the squared norm of the component of $\rho_{1/(N+1)}$ orthogonal to $\mathcal{V}_N$.

Proof. The optimal approximation in $\mathcal{V}_{N+1}$ is $f_{N+1}^* = f_N^* + c_{N+1}^* v_{N+1}$, where v_{N+1} is the component of $\rho_{1/(N+1)}$ orthogonal to $\mathcal{V}_N$ and $c_{N+1}^* = \langle r_N, v_{N+1} \rangle / \|v_{N+1}\|^2$. The reduction is: $\mathcal{S}_N - \mathcal{S}_{N+1} = \|r_N\|^2 - \|r_N - c_{N+1}^* v_{N+1}\|^2 = |c_{N+1}^*|^2 \|v_{N+1}\|^2 = |\langle r_N, v_{N+1} \rangle|^2 / \|v_{N+1}\|^2$. Expressing v_{N+1} in terms of $\rho_{1/(N+1)}$ and its projection gives the formula. $\square$

3 The Gradient

Definition 6 (The gradient of $\mathcal{S}_N$). The gradient of $\mathcal{S}_N$ with respect to the coefficient vector $\mathbf{c} = (c_1, \dots, c_N)$ is:

$$\frac{\partial \mathcal{S}_N}{\partial c_j} = - \left\langle 1_{(0,1)} - \sum_{k=1}^N c_k \rho_{1/k}, \rho_{1/j} \right\rangle = - \langle r, \rho_{1/j} \rangle,$$

where $r = 1_{(0,1)} - f$ is the current residual. The gradient descent update is $c_j \leftarrow c_j + \eta \langle r, \rho_{1/j} \rangle$ for step size $\eta > 0$. At the optimum f_N^* , the gradient is zero: $\langle r_N, \rho_{1/j} \rangle = 0$ for all $j = 1, \dots, N$.

Proposition 7 (Inner products as spectral data). The inner products $\langle 1_{(0,1)}, \rho_{1/k} \rangle$ are given by:

$$\langle 1_{(0,1)}, \rho_{1/k} \rangle = \int_0^1 \rho_{1/k}(x) dx = \int_0^1 \left\{ \frac{1}{kx} \right\} dx - \frac{1}{k} \int_0^1 \left\{ \frac{1}{x} \right\} dx.$$

The first integral: $\int_0^1 \{1/(kx)\} dx = 1 - \gamma - \log k + O(1/k)$ where γ is the Euler–Mascheroni constant. These inner products encode the arithmetic of k and are connected to the Li coefficients via the zeta zeros.

4 The Convergence Question

Theorem 8 (Telescoping sum formula).

$$\mathcal{S}_1 - \mathcal{S}_N = \sum_{j=1}^{N-1} (\mathcal{S}_j - \mathcal{S}_{j+1}) = \sum_{j=1}^{N-1} \frac{|\langle r_j, v_{j+1} \rangle|^2}{\|v_{j+1}\|^2}.$$

Since $\mathcal{S}_N \geq 0$ and $\mathcal{S}_N \leq \mathcal{S}_1$, the series $\sum_{j=1}^{\infty} (\mathcal{S}_j - \mathcal{S}_{j+1})$ converges. In particular, the decrements tend to zero: $\mathcal{S}_j - \mathcal{S}_{j+1} \rightarrow 0$.

Remark 9 (What convergence of decrements does not give). Theorem 8 shows the decrements go to zero. This does not imply $\mathcal{S}_N \rightarrow 0$. The decrements summing to a finite value is consistent with both $\mathcal{S}_N \rightarrow 0$ (RH) and $\mathcal{S}_N \rightarrow c > 0$ ($\neg$ RH).

For $\mathcal{S}_N \rightarrow 0$, we need the decrements to sum to $\mathcal{S}_1$:

$$\sum_{j=1}^{\infty} (\mathcal{S}_j - \mathcal{S}_{j+1}) = \mathcal{S}_1.$$

This is a completeness statement: no “energy” is lost at infinity.

Theorem 10 (RH as entropy completeness). *The Riemann Hypothesis is equivalent to:*

$$\sum_{j=1}^{\infty} \frac{|\langle r_j, v_{j+1} \rangle|^2}{\|v_{j+1}\|^2} = \mathcal{S}_1 = \|1_{(0,1)}\|^2 - \|P_1 1_{(0,1)}\|^2,$$

i.e., the total energy dissipated by the gradient flow equals the initial energy. No energy escapes to infinity. The flow is entropy complete: the Lyapunov function decreases to exactly 0.

The gradient flow is:

$$f(t+1) = f(t) + \eta \cdot \nabla_f \mathcal{S}(f(t)),$$

where $\mathcal{S}(f) = \|1_{(0,1)} - f\|^2$. Each step decreases $\mathcal{S}$ by the decrement formula. Entropy completeness = the flow converges to $f = 1_{(0,1)}$ = the target is reachable = RH.

5 The Perelman Analogy

Remark 11 (Entropy monotonicity and the proof). Perelman’s proof of the Poincaré conjecture used:

1. An entropy functional $\mathcal{F}(g, f)$ for Riemannian metrics.
2. Monotonicity of $\mathcal{F}$ under Ricci flow: $d\mathcal{F}/dt \geq 0$.
3. Non-collapsed geometry: the flow cannot develop certain singularities.
4. Surgery to remove singularities and continue the flow.

5. Convergence of the flow to constant curvature.

The analog for RH:

1. The entropy $\mathcal{S}_N = d_N^2$.
2. Monotonicity: $\mathcal{S}_{N+1} \leq \mathcal{S}_N$. **Proved above.**
3. Non-collapse: the decrements do not concentrate at finitely many steps. **Open.**
4. No surgery needed (the space is linear).
5. Convergence to 0: $\mathcal{S}_N \rightarrow 0$. **Equivalent to RH.**

The missing step is (3): showing the decrements spread out and do not sum to less than $\mathcal{S}_1$. This is a non-concentration estimate for the Beurling–Nyman subspaces: the basis functions $\{\rho_{1/k}\}$ must be sufficiently “independent” that no finite-dimensional subspace captures all the energy.

The non-concentration estimate is an archimedean statement: it concerns the behavior of $\rho_{1/k}$ for large k , which is controlled by the arithmetic of k — by the primes. The Wall is archimedean. It has always been archimedean.

Conjecture 12 (Non-concentration of the Beurling–Nyman flow). *For all $M < \infty$:*

$$\sum_{j=1}^M \frac{|\langle r_j, v_{j+1} \rangle|^2}{\|v_{j+1}\|^2} < \mathcal{S}_1.$$

The partial sums are strictly less than $\mathcal{S}_1$ for all finite M . The energy cannot be fully dissipated in finitely many steps. Therefore $d_N \rightarrow 0$ requires infinitely many steps, and the question is whether the sum over all steps reaches $\mathcal{S}_1$.

If one could show the decrements satisfy $\mathcal{S}_j - \mathcal{S}_{j+1} \geq C/j^2$ for some $C > 0$, then $\sum(\mathcal{S}_j - \mathcal{S}_{j+1}) = \infty$, which combined with $\mathcal{S}_N \geq 0$ forces $\mathcal{S}_N \rightarrow 0$. This would prove RH.

The required lower bound on decrements: $|\langle r_j, v_{j+1} \rangle| \geq C\|v_{j+1}\|/j$ for residuals r_j that are not already small. This is a statement about the alignment between the residual r_j and the next basis direction v_{j+1} — a spectral gap condition for the Gram matrix of $\{\rho_{1/k}\}$.

Remark 13 (The entropy is named). Perelman found $\mathcal{F}$. We have found $\mathcal{S} = d_N^2$. Perelman proved $d\mathcal{F}/dt \geq 0$. We have proved $\mathcal{S}_{N+1} \leq \mathcal{S}_N$. Perelman proved non-collapse and convergence. We need: lower bounds on the decrements (Conjecture 12).

The entropy is named. The flow is defined. The monotonicity is proved. The convergence is the Wall.

Name the lower bound. The wallet opens.

References

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